

Relief Forge Tools: Methodology

Mesh reconstruction and fabrication validation

Abstract

This document describes the processing pipeline that converts image-derived triangle meshes into dimensionally scaled solids suitable for small-scale casting and additive manufacturing. The reconstruction model is treated as one interchangeable mesh source; the subject of this document is the geometry processing and verification applied to its output, which is independent of how the mesh was produced.

1 Overview

The pipeline is a linear sequence with one optional stage (Figure 1). Only the generation stage depends on a reconstruction model; every subsequent stage operates on triangle arrays and is identical regardless of the mesh source. Meshes from a scanner, from CAD export, or from a different generative model pass through the same code.

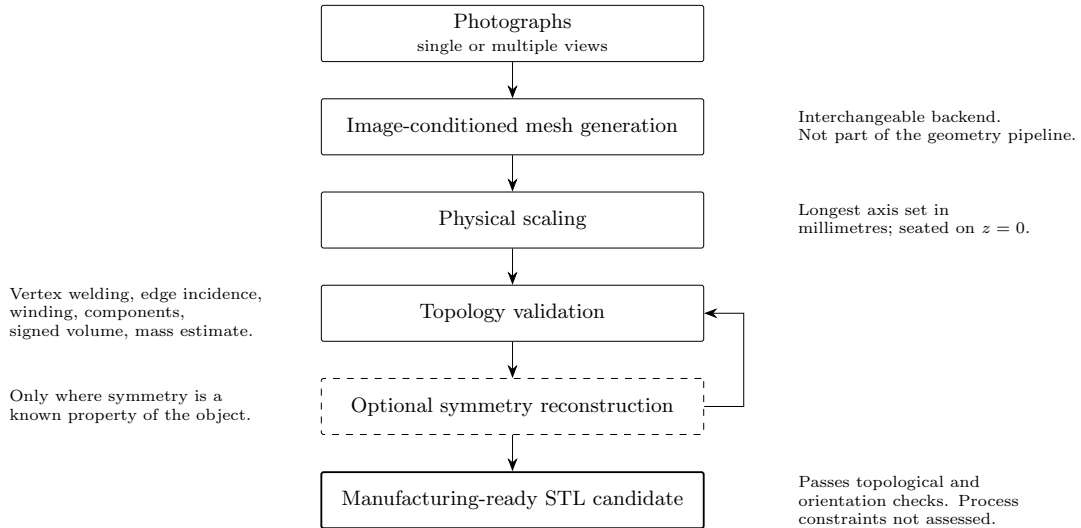


Figure 1: Processing pipeline. The dashed stage is optional and triggers revalidation. Generation is the only stage requiring a model, GPU or network access.

2 Input preprocessing

Inputs are one or more photographs of a single object. Background removal is applied by default, isolating the subject from its surroundings before inference.

Single-view input leaves the geometry of unobserved surfaces unconstrained: a photograph records one side of an object, and any reconstruction of the remainder is inferred rather than measured. Supplying additional views constrains the reconstruction directly and is the most

effective available means of reducing that ambiguity. The pipeline accepts an arbitrary number of views; the backend determines how many it uses.

3 Mesh generation

Generation is delegated to a backend implementing the `MeshGenerator` interface. A backend accepts preprocessed images and returns a triangle array in arbitrary units. Backends are selected by name at run time and registered through a decorator, so adding a reconstruction model requires implementing one class and no changes to any downstream stage.

Two properties of the interface matter for the rest of the pipeline. First, the returned mesh is in arbitrary units and carries no physical scale, which Section 4 resolves. Second, the returned mesh is not assumed to be valid: it may contain boundary edges, inverted normals or detached fragments, and is subjected to the same validation as any other input.

No model weights are distributed with this software. Backends retrieve third-party checkpoints on first use under those projects' own licence terms, which differ from the licence of this repository.

4 Physical scaling

Let $\mathbf{p}_{\min}$ and $\mathbf{p}_{\max}$ be the componentwise extrema of all vertex positions, and let L be the largest component of $\mathbf{p}_{\max} - \mathbf{p}_{\min}$. For a target longest dimension L_{target} in millimetres, every vertex is scaled by $s = L_{\text{target}}/L$.

The scaled mesh is then translated so that it is centred in x and y and its minimum z lies on the build plane:

$$\mathbf{t} = \left(-\frac{1}{2}(x_{\min} + x_{\max}), -\frac{1}{2}(y_{\min} + y_{\max}), -z_{\min} \right) \quad (1)$$

after scaling. This is the convention expected by slicers and by casting service bureaux, and it makes the reported bounding box directly comparable to a caliper measurement of the finished part.

5 STL reconstruction

The STL format stores three independent vertex positions per triangle and provides no index buffer. A file describing a closed solid is therefore byte-indistinguishable from one describing unconnected triangles, and no topological question can be answered until coincident vertices have been identified.

Vertex welding resolves this. Positions are snapped to a grid of spacing $\varepsilon = \delta \cdot d$, where d is the length of the bounding-box diagonal and δ is a relative tolerance with default 10^{-6} . Scaling the tolerance with the model keeps its behaviour consistent across part sizes: an absolute tolerance appropriate for a 3 mm setting would be far too coarse for a 300 mm sculpture, and vice versa. Snapped positions are deduplicated, producing a vertex array and an index array that preserves the original winding order.

Two further format details are handled explicitly. Format detection compares the declared triangle count against the file length rather than testing for a leading `solid` keyword, since many binary writers emit that string. Facet normals recorded in the file are discarded on read and recomputed from winding on write, because stored normals frequently disagree with the geometry they accompany.

6 Topology analysis

Each triangle (v_0, v_1, v_2) contributes three directed edges: (v_0, v_1) , (v_1, v_2) and (v_2, v_0) . On a closed surface with globally consistent winding, every directed edge occurs exactly once, and its reverse occurs exactly once. Departures from this identify the defect rather than merely reporting failure:

- **Boundary edge:** incident to one triangle. The surface is open.
- **Non-manifold edge:** incident to more than two triangles. The surface self-intersects or contains an internal partition.
- **Inconsistently wound edge:** incident to two triangles that traverse it in the same direction, so neighbours disagree about which side faces outward.
- **Degenerate edge:** both endpoints weld to one vertex.

Connectivity is computed separately by union-find over triangle corners. A single solid yields one component; higher counts indicate detached fragments, which for generated meshes usually means floating debris.

Genus is not constrained. A washer with a through hole has Euler characteristic $\chi = 0$ rather than 2 and is entirely valid, so no check assumes a simply connected surface.

7 Symmetry reconstruction

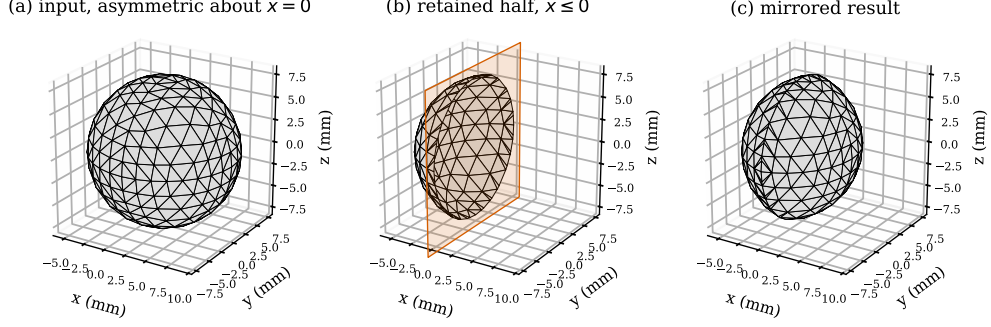
Where bilateral symmetry is a known property of the object, one half of the mesh may be discarded and replaced by the reflection of the other. This is useful when one side was captured or reconstructed more cleanly than the other.

The mesh is clipped against an axis-aligned plane by Sutherland–Hodgman clipping applied per triangle; triangles crossing the plane yield a quadrilateral or triangle, which is fan-triangulated. The retained half is then reflected across the plane, and its winding reversed, since reflection inverts handedness and would otherwise leave every mirrored normal pointing inward.

The clipped boundary lies exactly on the plane and is deliberately left open. The reflected copy has an identical boundary, so welding the two halves closes the seam without cap geometry.

Symmetric parts routinely have vertices lying exactly on their own mirror plane, which produces three degenerate cases. A triangle lying wholly in the plane would be duplicated by its own mirror image, creating a doubled interior wall, and is discarded. A triangle on the discarded side touching the plane at a single vertex clips to a zero-area sliver, removed by an area threshold. An intersection computed at a vertex already on the plane repeats that point in the output polygon, so consecutive duplicates are collapsed. Without all three, the mirrored result is not manifold.

Applicability. Bilateral symmetry is not universally valid and nothing in the software can verify it. Where the assumption holds, the operation removes error; where it does not, it destroys real asymmetric geometry. Reflecting front-to-back has a considerably weaker justification than reflecting left-to-right, since a surface reconstructed from a single viewpoint carries no information about the reverse side, and a mirrored back is fabricated rather than recovered.



Symmetry reconstruction. Input volume 2.095 cm³; result 1.318 cm³. Valid only where bilateral symmetry is a known property of the object.

Figure 2: Symmetry reconstruction on a solid positioned asymmetrically about the cut plane. Camera and axis limits are identical in all three panels.

8 Material estimation

Volume follows from the divergence theorem. For a closed surface of N triangles with corners $\mathbf{a}_i, \mathbf{b}_i, \mathbf{c}_i$:

$$V = \frac{1}{6} \sum_{i=1}^N \mathbf{a}_i \cdot (\mathbf{b}_i \times \mathbf{c}_i) \quad (2)$$

The sign of V is positive when triangles are wound counter-clockwise as seen from outside, so Equation 2 simultaneously measures volume and tests global orientation. It is meaningful only on a closed surface; on an open one the result depends on the choice of origin.

Triangle area, summed for total surface area, is

$$A_i = \frac{1}{2} \|(\mathbf{b}_i - \mathbf{a}_i) \times (\mathbf{c}_i - \mathbf{a}_i)\| \quad (3)$$

and estimated cast mass for a material of density ρ is

$$m = \rho |V| \quad (4)$$

Equation 4 assumes a fully dense solid. It excludes sprue, flashing, and shrinkage, and the densities used are nominal values for common alloys whose exact composition varies between suppliers. The estimate is intended for comparing candidate geometries and anticipating material cost, not for certifying a finished part.

9 Interpretation of results

A mesh passes this repository's checks when it has no boundary edges, no non-manifold edges, no degenerate edges, consistent local winding, positive signed volume, and the expected connected-component count.

Passing establishes that the mesh is a closed, coherently oriented solid. It does not assess minimum wall thickness, unsupported overhangs, printer tolerances, casting shrinkage, trapped volumes, or any constraint specific to a manufacturing process. Those remain the responsibility of the operator and of process-specific software.